AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW

RESOURCE LOCAL LANGUAGES

R26-DS-11



Meet Our Team



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Problem in Brief

The Problem

- During disasters, emergency centres receive many calls and SMS messages.
- People may use Sinhala, Tamil, English, or mixed languages.

Operator Challenge

- Callers may not explain the emergency clearly or give all important details.
- Operators must understand the emergency, identify urgency, and check for duplicate reports.

Main Pain Point

These tasks take time and can delay emergency response.

Our goal is to help operators understand emergency information faster.



Many reports | Missing details | Time pressure

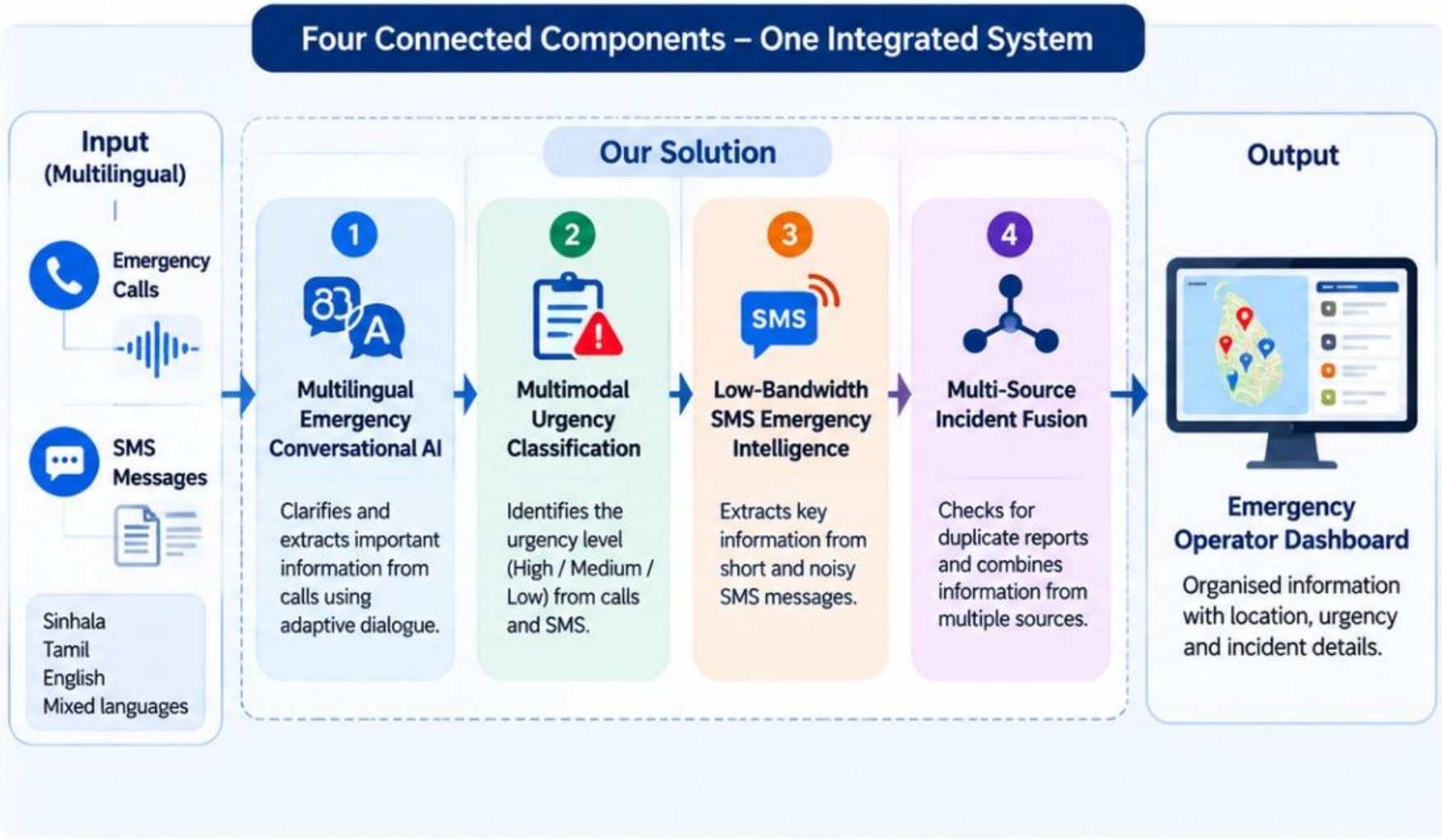
Overall Design Excellence

1. Our solution has **four connected components** that work together as one system.

2. The system **processes** emergency calls and SMS messages, **collects important information**, **identifies urgency**, and **checks for duplicate reports**.

3. Finally, the information is organised and displayed on the **emergency operator dashboard**.

4. Our main contribution is bringing these four components together to support **faster and better emergency response** in **Sri Lanka**.



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Multilingual Emergency Conversational AI System For Rapid Clarification



Multilingual Emergency Clarification

- Identifies important emergency details from the caller's speech.
- Finds missing information.
- Asks follow-up questions in the caller's language (Sinhala, Tamil, English or mixed).
- Adapts questions based on the conversation and caller's situation.

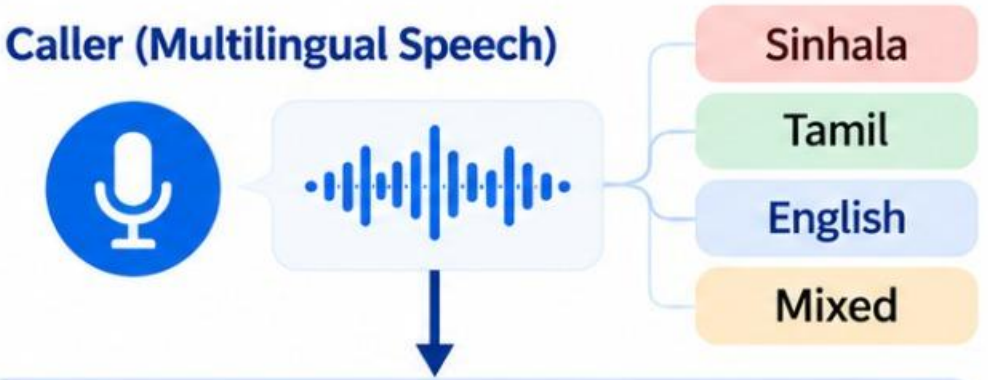
Example Clarification Conversation



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Multimodal Urgency Detection Using Prosodic and Conversational Cues

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Urgency Classification

- Uses speech content and voice information (tone, emotion, and keywords).
- Classifies emergencies as **High, Medium,** or **Low urgency.**
- Helps operators to prioritise urgent incidents.
- Works for Sinhala, Tamil, English and mixed languages.

Input Features

- Speech Content (transcript)
- Voice Information (tone, emotion)
- Multilingual Support

Classification

- **High Urgency** — Immediate response needed
- **Medium Urgency** — Response required
- **Low Urgency** — Monitor and plan

Example Classification



Caller Speech

→ **Urgency Classification Model** →

- High Urgency
- Medium Urgency
- Low Urgency

Low-Bandwidth SMS Emergency Intelligence for Disaster Response

IT22263476 | D.M.S.G.Karunarathna

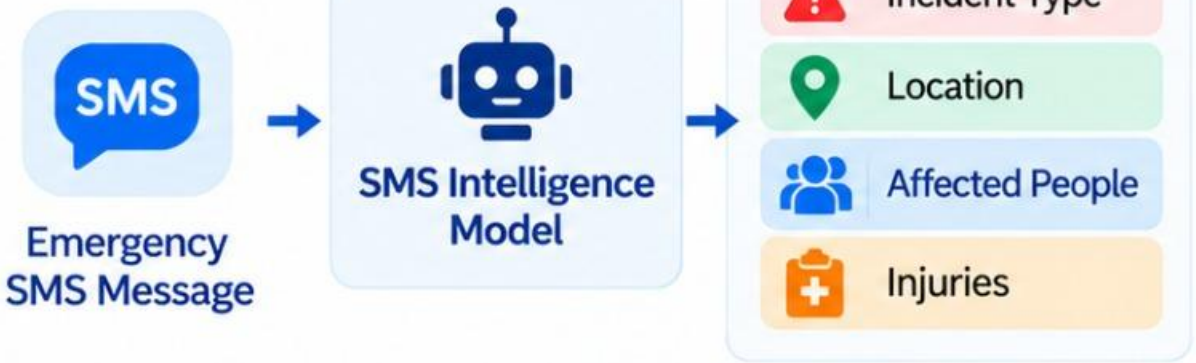
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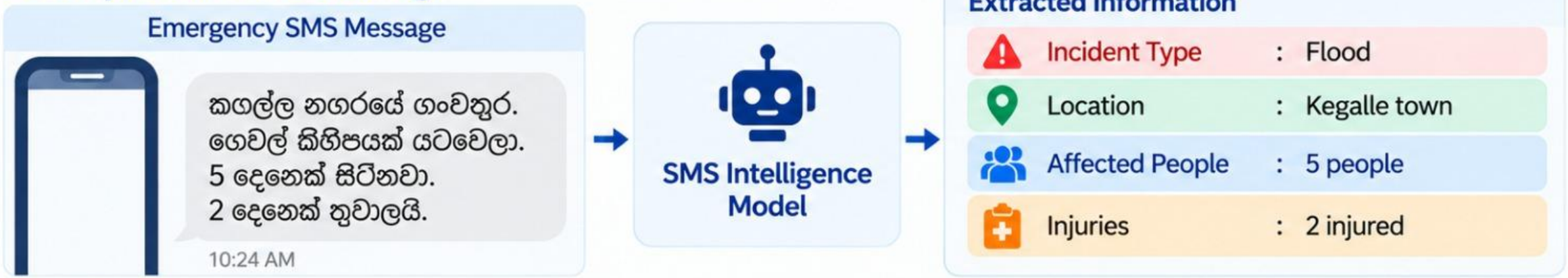
Low-Bandwidth SMS Intelligence

- Processes emergency SMS messages from the public.
- Extracts important information such as incident type, location, affected people and injuries.
- Works with short, simple and low-bandwidth messages.
- Supports Sinhala, Tamil, English and mixed languages.

SMS Processing Flow



Example SMS Processing



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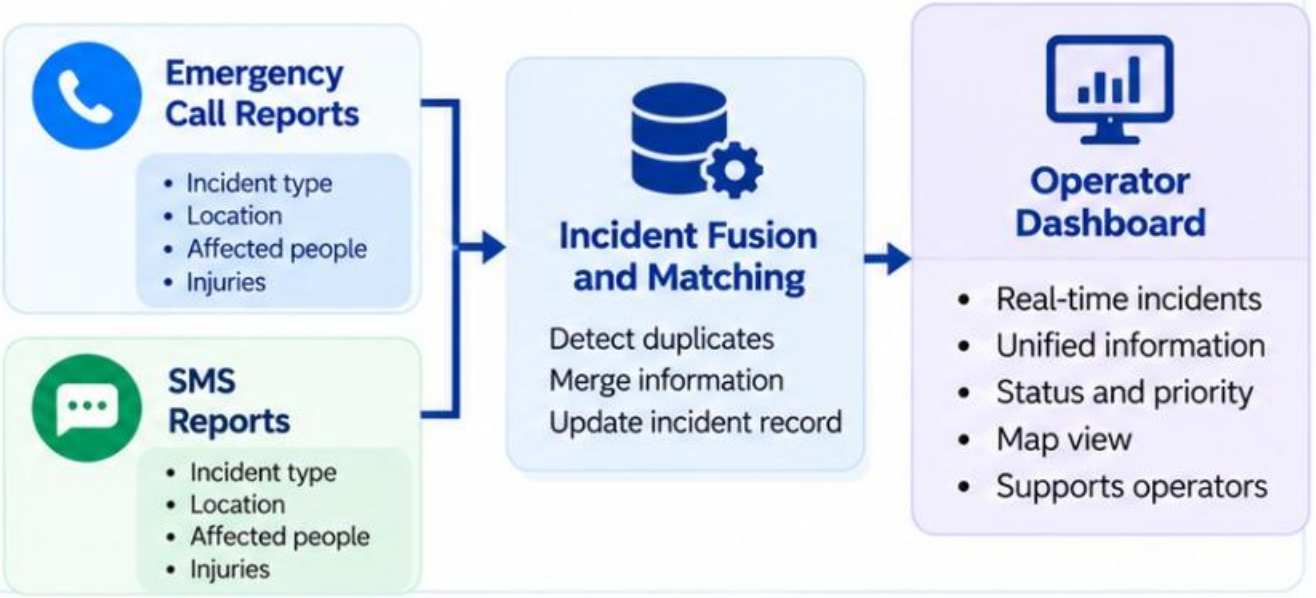


Multi-Source Urgency Based Incident Fusion ,UIRs & Real - Time Operator Dashboard

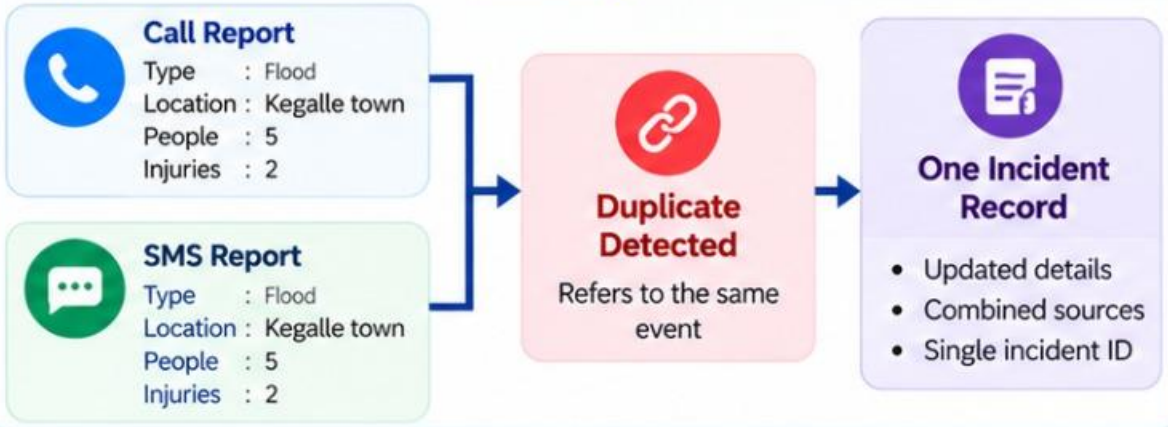
Incident Fusion and Operator Dashboard

- Combines reports from emergency calls and SMS messages.
- Identifies possible duplicate incidents (refers to the same event).
- Merges and updates information from different sources into one incident record.
- Displays incidents in a real-time operator dashboard.
- Helps operators to get a complete and accurate situation for faster response.

Incident Fusion Process



Duplicate Incident Matching



Operator Dashboard (Example)



ID	Incident Type	Location	Source	Priority	Status
INC-001	Flood	Kegalle town		High	Dispatched
INC-002	Landslide	Ratnapura		Medium	In Progress
INC-003	Fire	Colombo		High	New
INC-004	Flood	Matara		Low	Verified

Results, Feedback and Limitations

Results of the Four Components

Clarification Component

- Identifies missing details.
- Asks relevant follow-up questions in multiple languages.

Urgency Component

- Classifies emergency priority as High, Medium or Low.
- Uses speech and text information.

SMS Component

- Extracts important information from SMS messages.
- Organises location, incident type, affected people and injuries.

Fusion Component

- Identifies duplicate reports.
- Merges information from calls and SMS into one incident record.
- Displays organised data for operators.

Prototype Feedback

- The prototype feedback was positive.
- Users found the system useful for handling emergency information.
- It can support operators and help them work faster.

Limitations

- Limited real-world emergency data for evaluation.
- Need better support for local languages (Sinhala, Tamil) and more mixed-language data.
- Further testing required under noisy conditions.
- Integration of all components needs improvement before full real-time deployment.

Target Users

Disaster Management Centre (DMC)

Main target user

- Police (119)
- Rescue Teams
- Provincial Units
- Other Emergency Services (Future Support)

Conclusion

Our project aims to make **emergency information clearer**, help **operators work faster**, and support **better disaster response** in Sri Lanka.